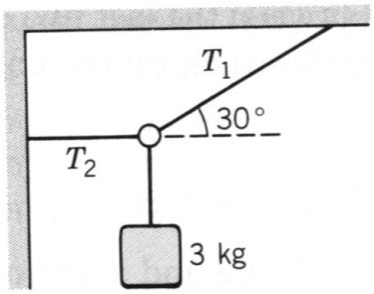
Homework #6

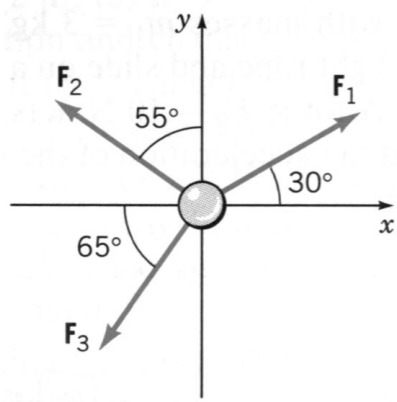
1) Compute the acceleration for the following in terms of g's where g = 9.81 m/s^2 : (a) a car moving at 100. km/h round a curve of radius 50.0 m; (b) a jet flying at 1.50×10^3 km/h and making a turn of radius 5.00 km; (c) a stone being twirled every 0.500 s at the end of a rope of length 1.50 m; (d) a speck of dust at the rim of an LP with a radius of 6.00 in. turning at 33.1 rpm; (e) a molecule in a centrifuge rotating at 30,000 rpm at a radius of 15.0 cm.

2) Suppose the earth's rotation speed up such that the centripetal acceleration at the equator equaled g. What would be the length of a "day"?

3) A 3.00-kg block is suspended with two ropes, one of which is horizontal, as shown in the figure. Find the tension in each rope.

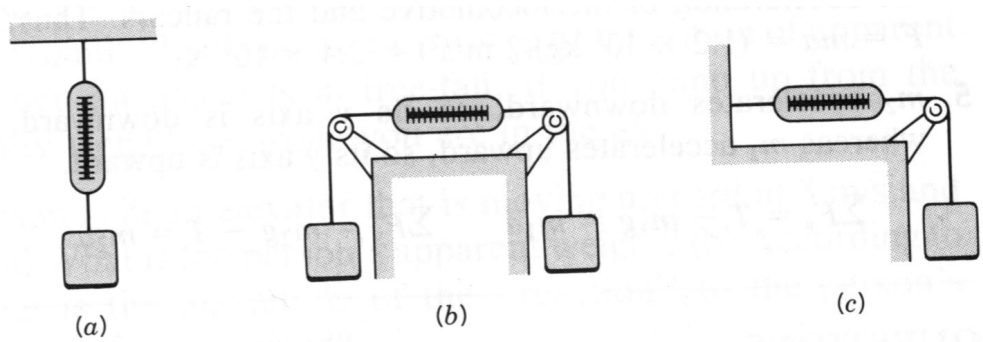


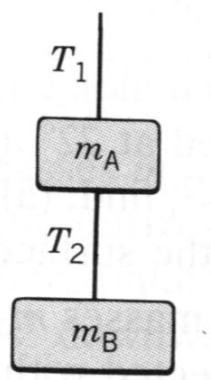
4) Three forces with magnitudes $F_1 = 12.0$ N, $F_2 = 16.0$ N, and $F_3 = 7.00$ act on a 0.450 kg body in the directions shown in the figure below. What is the acceleration in unit vector notation?



5) What is the constant force required to accelerate a 1225 kg car in each of the following cases: (a) It starts from rest and reaches 96.0 km/h in 10.0 s; (b) it is braked to a stop from 112 km/h in 64.0 m. (c) In each case, what is the origin of the force?

6) What is the reading on the spring scale for each of the situations depicted in the figure below? Each of the blocks has a weight of 12.0 N.



7) A rocket has a mass of 1.40×10^4 kg and its engine has a thrust of 2.00×10^5 N. If its engines fire in the vertical direction for 1 min starting from rest, to what vertical height would it rise in the absence of air resistance? (*To simplify things, assume the mass is constant. However, in reality the mass of rockets decreases substantially while rising*)

8) Two blocks with masses $m_A = 0.200$ kg and $m_B = 0.300$ kg hang one under the other, as shown in the figure below. Find the tensions in the (massless) ropes in the following situations: (a) The blocks are at rest; (b) they move upward at 5 m/s; (c) they accelerate upward at 2.00 m/s^2; (d) they accelerate downward at 2.00 m/s^2. (e) If the maximum allowable tension is 10.0 N, what is the maximum possible upward acceleration?